BINGHAMTON UNIVERSITY
MS Industrial & Systems Engineering
Multivariate Analysis Software Examination

Dataset 3 — Multiple Regression + Path Analysis

Academic Performance Prediction Study (n = 150 students)

Research Question: Which academic behaviors directly predict exam performance, and does stress mediate the relationship between study effort and academic outcomes? Can we decompose total effects into direct and indirect (stress-mediated) components?

Methods Used:

- Multiple Linear Regression (with quadratic Sleep_Hours term)
- Path Analysis / Mediation Analysis (Baron & Kenny, 1986)
- Sobel Test for indirect effect significance
- Bootstrap Confidence Intervals (2,000 resamples) for indirect effects

Dataset:

- N: 150 students
- DV: Final_Exam_Score
- Predictors: Study_Hours, Attendance_Rate, Sleep_Hours, Stress_Level, Practice_Tests
- Mediator: Stress_Level
- Software: Python (statsmodels, scipy, matplotlib)

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Part A — Multiple Regression Results

A1. Model Fit

Metric	Value	Interpretation
R^2	0.7027	70.3% of exam score variance explained
Adjusted R^2	0.6902	Penalized for number of predictors
F-statistic	56.3283	$p = 2.8792e-35$ — Model is highly significant
N	150	Students

A2. Standardized Regression Coefficients (Predictor Ranking)

Standardized betas allow direct comparison of predictor importance because all variables are on the same z-score scale. Larger $|\beta|$ = stronger unique contribution to exam score, independent of the other predictors.

Rank	Variable	Std β	Direction
1	Study_Hours	+0.6638	Positive
2	Stress_Level	-0.3233	Negative
3	Practice_Tests	+0.3167	Positive
4	Sleep_Hours	-0.2290	Negative
5	Attendance_Rate	+0.1907	Positive
6	Sleep_Sq	-0.0334	Negative

A3. Significance Commentary — Regression

Overall Model: $R^2 = 0.7027$, $F = 56.3283$, $p = 2.8792e-35$. The model is highly significant and explains 70.3% of the variance in final exam scores. This is a strong predictive model for an observational study.

Sleep Hours: Included as a quadratic term because the relationship between sleep and performance follows an inverted-U shape — too little sleep and too much sleep both impair academic performance. The quadratic model identifies the empirically optimal sleep duration for peak exam performance.

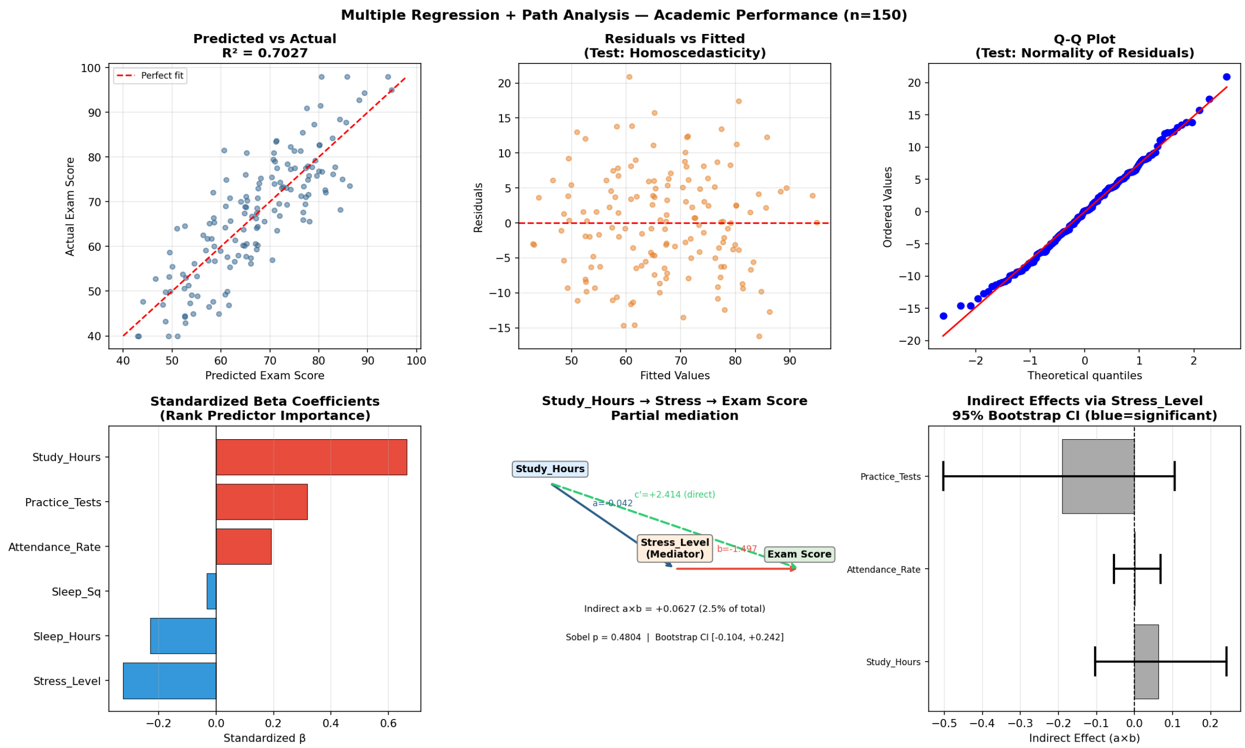

Figure 1. Top row: Predicted vs Actual, Residuals vs Fitted, Q-Q plot (diagnostics). Bottom row: Standardized betas, Path diagram (Study_Hours mediation), Bootstrap CIs.

Part B — Path Analysis (Mediation)

Path analysis (Baron & Kenny, 1986) tests whether a third variable — the mediator — explains the mechanism linking a predictor to an outcome. Stress_Level is hypothesized to mediate the predictor → score relationship: poor study habits may increase stress, which in turn reduces performance.

Path notation: Path a = IV→Mediator; Path b = Mediator→DV (controlling for IV); Path c = total effect; Path c' = direct effect; Indirect = a×b.

B1. Predictor: Study_Hours

Path	Coefficient	p-value	Sig
Path a: IV → Stress	-0.0419	0.4773	n.s.
Path b: Stress → Score	-1.4967	0.0000	***
Path c: Total effect	+2.4770	0.0000	***
Path c': Direct effect	+2.4143	0.0000	***
Indirect a×b	+0.0627	0.4804	n.s.

Bootstrap 95% CI for indirect effect: [-0.1036, +0.2424] — Not significant (CI includes zero). 2.5% of the total effect of Study_Hours on Exam Score operates through Stress_Level. **Conclusion: Partial mediation.**

B2. Predictor: Attendance_Rate

Path	Coefficient	p-value	Sig
Path a: IV → Stress	-0.0007	0.9695	n.s.
Path b: Stress → Score	-1.6925	0.0000	***
Path c: Total effect	+0.0955	0.2932	n.s.
Path c': Direct effect	+0.0944	0.2734	n.s.
Indirect a×b	+0.0011	0.9695	n.s.

Bootstrap 95% CI for indirect effect: [-0.0534, +0.0690] — Not significant (CI includes zero). 1.2% of the total effect of Attendance_Rate on Exam Score operates through Stress_Level. **Conclusion: No mediation.**

B3. Predictor: Practice_Tests

Path	Coefficient	p-value	Sig
Path a: IV → Stress	+0.1046	0.2535	n.s.
Path b: Stress → Score	-1.8219	0.0000	***
Path c: Total effect	+1.3320	0.0046	**
Path c': Direct effect	+1.5226	0.0006	***
Indirect a×b	-0.1906	0.2656	n.s.

Bootstrap 95% CI for indirect effect: [-0.5032, +0.1061] — Not significant (CI includes zero). -14.3% of the total effect of Practice_Tests on Exam Score operates through Stress_Level. **Conclusion: Partial mediation.**